# Chatbot for college inquiry system

# Objective:

The objective of this project is to build an intelligent chatbot for answering college-related questions.

It aims to automate responses for common inquiries like admissions, fees, courses, and scholarships.

**Aim:**

To develop a chatbot that helps prospective or current students get accurate and human-like responses for frequently asked college questions related to:

- Admissions
- Courses and curriculum
- Fees and scholarships
- Hostel facilities
- Placements and internships
- College infrastructure and facilities

## Dataset Description (Related Data)

**📁 Filename:**

college_qa_dataset.csv

**📄 Columns:**

Column Name Description

input A list of common student questions (e.g., "What are the hostel facilities?")

# Models Used:

## 1. Sentence Transformer (all miniLm-l6-v2)

**Sentence Encoding:**

It converts a sentence into a vector (embedding) that captures its meaning.

**Semantic Understanding:**

Understands the context and meaning of a sentence,

**Maintains Accuracy:**

MiniLM delivers competitive accuracy on many NLP tasks.

## Flan-T5

**Generates human-like answers :**

when the user’s question doesn't closely match any FAQ.

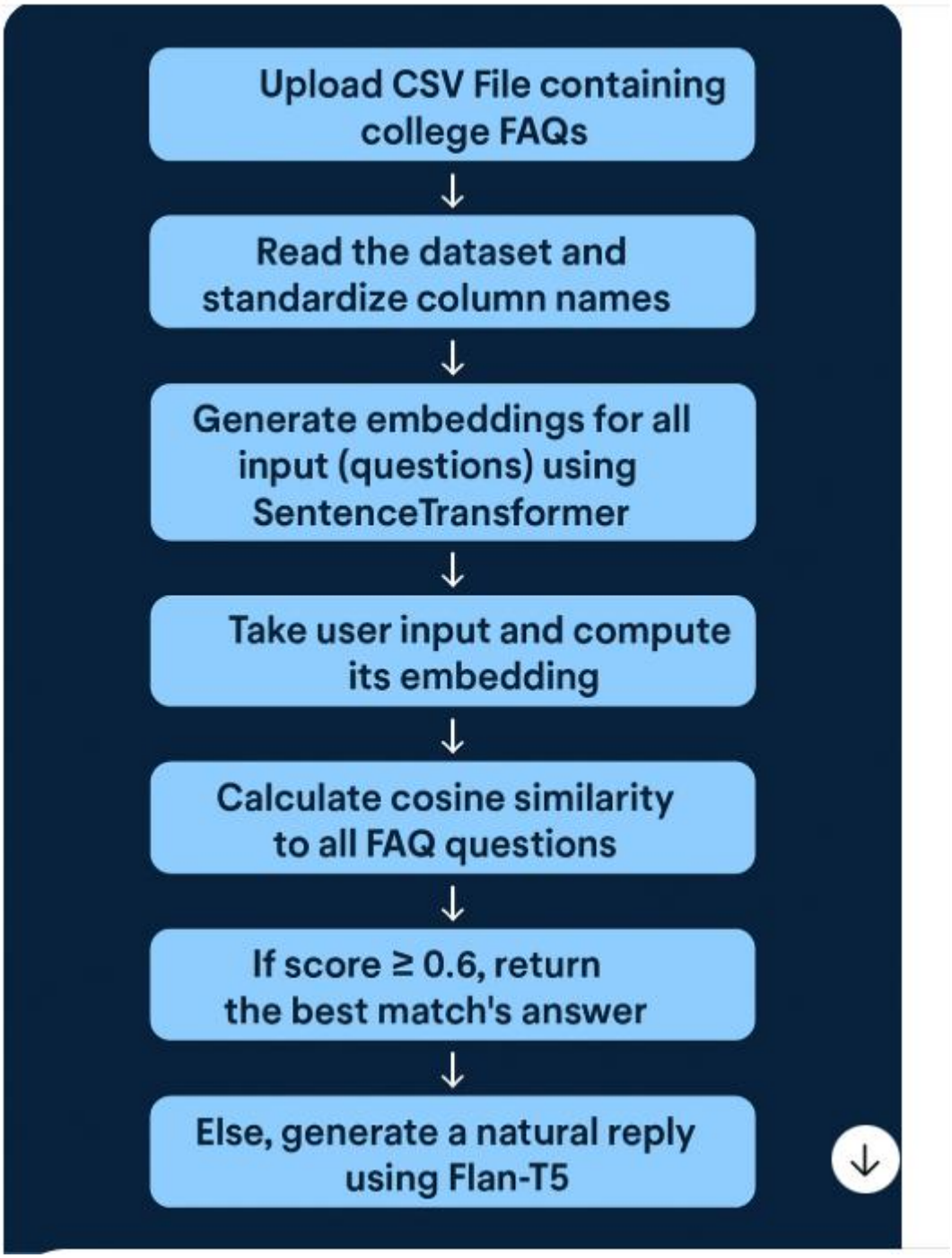
**Text Generation:**

Generates relevant replies by understanding the meaning of the question.

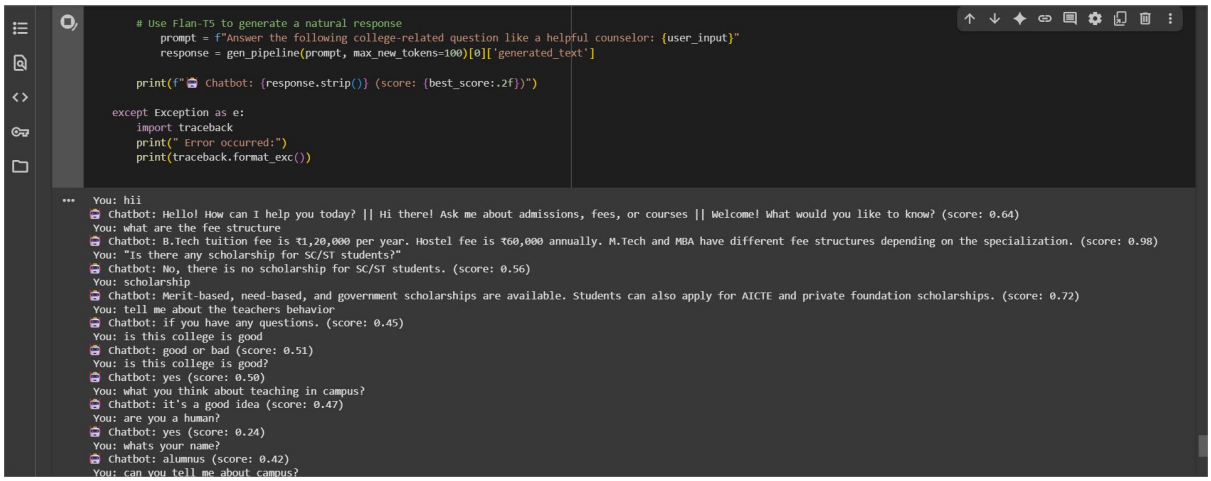
**Instruction Following:**

Responds based on specific prompts like “Answer like a helpful counselor.”

# Diagram:


Upload CSV File containing college FAQs
Read the dataset and standardize column names
Generate embeddings for all input (questions) using SentenceTransformer
Take user input and compute its embedding
Calculate cosine similarity to all FAQ questions
If score ≥ 0.6, return the best match's answer
Else, generate a natural reply using Flan-T5

# Output:



# Use Flan-T5 to generate a natural response
prompt = f"Answer the following college-related question like a helpful counselor: {user_input}"
response = gen_pipeline(prompt, max_new_tokens=100)[0]['generated_text']
print(f" Chatbot: {response.strip()} (score: {best_score:.2f})")
except Exception as e:
import traceback
print(" Error occurred:")
print(traceback.format_exc())
You: hii
Chatbot: Hello! How can I help you today? || Hi there! Ask me about admissions, fees, or courses || Welcome! What would you like to know? (score: 0.64)
You: what are the fee structure
Chatbot: B.Tech tuition fee is ₹1,20,000 per year. Hostel fee is ₹60,000 annually. M.Tech and MBA have different fee structures depending on the specialization. (score: 0.98)
You: "Is there any scholarship for SC/ST students?"
Chatbot: No, there is no scholarship for SC/ST students. (score: 0.56)
You: scholarship
Chatbot: Merit-based, need-based, and government scholarships are available. Students can also apply for AICTE and private foundation scholarships. (score: 0.72)
You: tell me about the teachers behavior
Chatbot: if you have any questions. (score: 0.45)
You: is this college is good
Chatbot: good or bad (score: 0.51)
You: is this college is good?
Chatbot: yes (score: 0.50)
You: what you think about teaching in campus?
Chatbot: it's a good idea (score: 0.47)
You: are you a human?
Chatbot: yes (score: 0.24)
You: whats your name?
Chatbot: alumnus (score: 0.42)
You: can you tell me about campus?